Informative Path Planning in AUV retrieval

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***Abstract*—This paper presents a probabilistic framework for addressing a Coverage Path Planning (CPP) problem. Given a GPS measurement from an Autonomous Underwater Vehicle (AUV), the objective is to generate a path for an Unmanned Aerial Vehicle (UAV) that accounts for measurement uncertainty, AUV motion induced by environmental factors, and potential failures in the AUV's perception subsystem. Two informative planning strategies (a greedy approach and an Artificial Potential Field (APF)-based method) are proposed and compared against a spiral baseline. The results show that while the spiral planner yields the shortest flight times under low-noise conditions, the APF method achieves slightly lower flight times when GPS measurements are strongly degraded. Moreover, with the chosen model hyperparameters, it maintains coverage performance close to that of the spiral planner while exhibiting greater flexibility than the greedy approach.**

***Index Terms*—Unmanned Aerial Vehicles, Path Planning, Bayes Filter**

I. Introduction

Modern autonomous underwater vehicle (AUV) operations play a critical role in naval intelligence, surveillance, reconnaissance, environmental monitoring, and subsea exploration [1]. However, traditional deployment and recovery methods, which rely on mothership-based handling, suffer from operational inefficiencies and safety concerns.

To address these limitations, the ALARS project (Aerial Launch and Recovery System for Autonomous Underwater Vehicles), which was funded by SAAB AB, Digital Futures, and Vinnova (DNR 2024-01576), was developed. ALARS integrates unmanned aerial vehicles (UAVs, more specifically, quadrotors) into the deployment and recovery of AUVs, improving operational efficiency, increasing flexibility, and reducing risk to human operators. The recovery process can be broadly divided into three main phases: search, detection, and retrieval [2]. Although these phases are tightly interdependent and ideally operate seamlessly, this paper focuses exclusively on the search phase, which is critical for minimising the time required to locate the AUV and has a direct impact on the overall mission success.

In the ALARS mission setup, a DJI Matrice 350 RTK drone is equipped with a NVIDIA Jetson Nano and multiple sensing systems, including an IMU and an RGB camera. Its objective is to locate the SAM (Small & Affordable Maritime Underwater Robot), which remains attached to a buoy to increase the likelihood of successful detection. At the start of each mission, a GPS (Global Positioning System) measurement—hereafter referred to as a ping—provides an initial estimate of the AUV's position, although this estimate is subject to uncertainty. The environment is assumed to be free of obstacles, which makes a spiral motion planner a natural baseline candidate to use in this Coverage Path Planning (CPP) problem.

The objective of this work is to evaluate whether it is possible to maintain search times (in comparison to a spiral planner) while improving robustness against false detections. Underwater environments introduce challenges such as reflections, environmental disturbances, and sudden vehicle motions, which can degrade detection performance and increase the likelihood of false positives and negatives. Additionally, waves and wind can induce unpredictable AUV drift over time, requiring planners to account for dynamic position changes. Although the spiral planner is efficient, it does not support multiple traversals of the same region without significantly reducing coverage efficiency.

To address these challenges, the following approaches were compared:

1) a spiral motion planner, serving as a baseline;
2) an informative path planner that models the AUV's presence within a probabilistic framework, for which two algorithms were developed: a greedy strategy and one based on an Artificial Potential Field.

All experiments were conducted in the Unity simulator developed within the Robotics, Perception, and Learning (RPL) division at KTH [3].

II. Related Work

The CPP problem has been extensively studied over the years, driven by the growing interest in UAV-based solutions for surveillance, agriculture, and search-and-rescue operations [4]. However, its formulation can be quite case-specific, since different requirements and constraints heavily change how one addresses the problem. According to [5], CPP methods can be broadly categorized into classical and heuristic-based algorithms. The latter include greedy or graph-search methods and bio-inspired approaches. Greedy and graph-search algorithms typically offer low search times, whereas bio-inspired methods tend to achieve more optimal or near-optimal coverage. Even so, classical approaches generally exhibit lower computational complexity, which is crucial in hardware-constrained applications.

This problem lies within the family of CPP methods that exploit prior information about the environment or target. As mentioned earlier, selecting an appropriate CPP strategy depends strongly on the available knowledge. For instance,

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a boustrophedon pattern [4] may be appropriate for completely unknown areas, whereas access to localisation cues (e.g., a GPS ping) can make spiral trajectories [6] or other information-driven paths more effective. At the same time, the task is closely related to probabilistic search and Search and Rescue (SAR) robotics, as it requires covering the relevant region while maximising the probability of locating the AUV under uncertainty.

Bourgault *et al.* [7] proposed a decentralized Bayesian method for coordinating multiple autonomous sensor platforms tasked with locating a single, non-evading target. Their formulation explicitly models the vehicles' kinematics, the sensors' detection characteristics, and the target's motion model. Although a quasi-optimal solution is defined for each platform through a utility function based on cumulative detection probability, the primary contribution of the work is the coordinated planning mechanism. This coordination is achieved using a Bayesian network through which agents exchange information, allowing each decision maker's beliefs (and consequently its subsequent decisions) to be influenced by the updated priors provided by other agents.

The relationship between information availability and path planning efficiency is explored in [8], where Gaussian distributions combined with a Naive Bayes filter are used to update the belief map. The study focuses on a search-and-rescue scenario in which a UAV must locate a missing person or detect a potential fire within a designated area. A multi-peaked Gaussian prior is employed, but the Bayesian filtering formulation does not clearly separate the prediction and update steps, likely due to the static nature of the target. Nonetheless, the results show that informative path planning can substantially improve efficiency, achieving an 80% reduction in mean flight time compared to a standard zigzag strategy.

Andersson *et al.* [9] propose a novel framework that combines approximate Bayesian learning via INLA (Integrated Nested Laplace Approximation) with a tailored MCTS (Monte Carlo Tree Search) motion-planning strategy for search tasks. In their formulation, population intensity, detection probability, and injury probability are modeled and fed into INLA. The authors evaluate their method against a conventional "lawnmower" motion planner across several simulated scenarios. Their results indicate that leveraging prior information significantly improves search performance, with the INLA-MCTS approach outperforming the baseline.

While the CPP problem without energy constraints has been extensively studied in the literature (e.g., RRT and spiral coverage patterns), the energy-constrained variant has recently received increasing attention. This formulation involves defining an appropriate energy model and representing the environment as either a graph or a polygon. In [10], the authors introduce the concept of an equi-distance-contour-connected environment for robots constrained to axis-parallel motion. Their approach is later generalised to arbitrary polygons by partitioning the workspace into contour-connected regions and defining the energy function as the distance traveled.

In another line of work, [11] propose an optimal CPP planner based on the Iterative Deepening Depth-First Search (ID-DFS) algorithm. Since the method exhibits worst-case exponential time complexity, the authors introduce two improvements (loop detection and an admissible heuristic function) that preserve optimality while achieving orders-of-magnitude faster performance compared to exhaustive search.

Conversely, offline strategies can be employed [12]. Using an approximate cellular decomposition and a wavefront algorithm, an optimal criterion based on both route length and the number of turns is applied for path planning. After evaluating all possible paths and selecting the one that minimizes the cost, the resulting trajectory is then smoothed using a cubic spline.

While previous approaches have yielded promising results, the scenario considered in this work differs in several key aspects. The environment is obstacle-free and centered around a known but uncertain GPS ping, which defines a single region of interest rather than requiring full-area coverage. Moreover, only a few existing studies explicitly address the combined effects of GPS uncertainty and dynamic AUV drift on search performance. Under these conditions, a spiral motion planner emerges as a natural baseline, offering efficient coverage when the target location is approximately known. The primary objective of this work is therefore to develop and evaluate alternative planners that can mitigate false detections and maintain comparable time of flight (ToF) performance relative to the spiral approach. Furthermore, given the hardware constraints, a lightweight search model is preferred, commensurate with the task complexity.

III. Methodology

A. Spiral algorithm

The simplest and most traditional algorithm for CPP problems. This algorithm actually contains three phases: a straight line to the GPS ping, a circular motion around it, and finally the moving spiral. This choice stems from the need to maximise efficiency by using UAV's FOV (Field of Vision) while ensuring there's no missing area. The spiral points (x_k, y_k) were parameterised as follows in (1):

$$\begin{cases} x_{i+1} = x_{GPS} + r_i cos(\theta_i) + kTv_{x,AUV} \\ y_{i+1} = y_{GPS} + r_i sin(\theta_i) + kTv_{y,AUV} \\ r_{i+1} = r_i + h\frac{1}{N}tan(\frac{FOV}{2})(1 - \sqrt{\frac{|v|}{|v_{max}|}}) \end{cases} \quad (1)$$

Here, r_i is the spiral radius at time step i, T is the predicted period of the AUV, h is the flight altitude, N is the number of waypoints in the spiral, and v denotes the AUV's velocity. The velocity scaling factor k is set to 2 to ensure that the spiral center moves faster than the AUV. This choice may introduce some discontinuities, but it guarantees that no area is left uncovered. This parameterization models a dynamic spiral that follows the AUV, at the cost of coverage. As the AUV's velocity increases, the coverage radius decreases, as shown in (1). In Fig. 1, the path planner result is shown for different AUV velocities, as well as the spiral center for each spiral revolution.

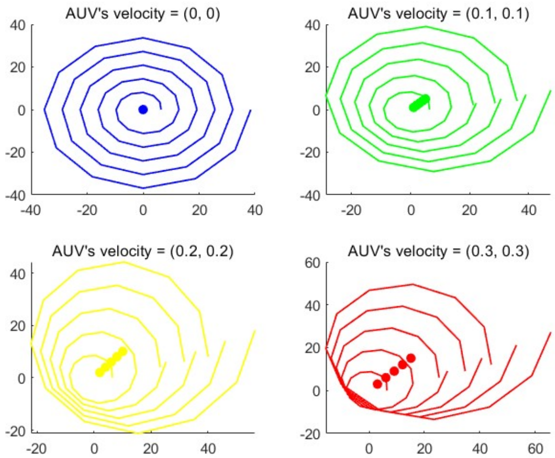

Fig. 1: Spiral planner for different AUV's velocities, where $|v_{max}|$ was set to 2

B. Informative algorithms: probabilistic framework

These algorithms make use of a Probabilistic Grid Map (PGM), which consists of a rectangular grid map with a probability assigned to each cell. The map is constantly updated with a Bayes Filter prediction and update steps [7]. While the update step is mathematically well-defined, the prediction step lacks a consistent dynamic model $p(x_{t+1} \mid x_t)$, since the AUV's motion depends on external factors such as wind and waves, and it's not that trivial to convert them to a probability in a discretized map. To address this simply, an approximation is employed: a convolution is applied to transform the AUV's velocity into probabilities, by encoding an approximate dynamic model in the filter/kernel values.

Prediction

The probabilities are predicted with a convolution with a kernel that takes the estimated AUV's velocity into account, which is described in (2):

$$\mathbf{P}(x_{t+1}|z_1, ..., z_t) \propto \mathbf{P}(x_t|z_1, ..., z_t) \circledast \mathbf{K} \tag{2}$$

Where x_t represents the "AUV in cell at time t," x_t is the binary detection measurement, $\mathbf{P}$ denotes the probabilistic graphical model in matrix form, and $\mathbf{K}$ is a 3×3 filter whose weights are aligned with the estimated velocity vector of the AUV. For example, if the AUV is moving along the y-axis:

$$\mathbf{K} = \begin{bmatrix} q \times k_{blur} & q & q \times k_{blur} \\ 0 & Q & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{3}$$

Where Q denotes the central weight, q the non-central weight, and k_{blur} a blur factor introduced in the 4-neighbors cells (apart from the central element) to account for the error in the AUV's velocity vector. This means we have eight possible kernel configurations, *ie*, eight possible velocity directions. Thus, the estimated AUV's velocity has to be mapped to one of these directions beforehand, resulting in precision loss. One could use a bigger kernel, but that would entail more degrees of freedom (more weights).

Since the probabilities are normalised at each step, only the ratio between the weights is required and is computed as follows:

$$\frac{q}{Q} = k_{\text{weight}} \cdot \frac{d}{d_{\max}}, \qquad k_{\text{weight}} = \begin{cases} 10^{-8}, & \text{if } \dfrac{d}{d_{\max}} < 10^{-3} \\ 0.1, & \text{otherwise.} \end{cases} \tag{4}$$

Where d is the AUV's distance travelled between prediction steps, d_{max} corresponds to the diagonal of a single cell. By using a piecewise formulation, abrupt variations in the PGM due to measurement noise are avoided, resulting in a more stable representation. The parameters k_{blur} and k_{weight} were empirically fine-tuned across different scenarios to ensure that the most likely cell preserved a velocity comparable to that of the AUV.

Update

The Bayes Filter update step can be formulated as:

$$p(x_t \mid z_{1:t}) \propto \underbrace{p(z_t \mid x_t)}_{\text{likelihood}} \underbrace{p(x_t \mid z_1, ..., z_{1:t-1})}_{\text{prior}} \tag{5}$$

Given its recursive nature, the prior only needs to be explicitly defined at the initial step, which can be done with a Gaussian distribution centred at the GPS ping with a variable variance σ_P. As for the likelihood, let β denote the probability of a false negative. Thus, we can define the likelihood [7] as follows in (6).

$$p(z_t \mid x_t) = \begin{cases} \beta, & \text{if not } z_t \\ 1-\beta, & \text{otherwise} \end{cases} \tag{6}$$

One could use a more robust sensor model to include, for instance, the probability of having a false positive [8]. However, that would involve including the posterior $p(z_t \mid \overline{x_t})$ in the PGM formulation.

It's worth noting that while the predict step is applied to the entire PGM, the (unnormalized) update step is applied only to the "seen" cells, which correspond to the cells enclosed by a circle centered at the UAV's position and whose radius is given by

$$r = h \times tan(\frac{FOV}{2}) \tag{7}$$

C. Greedy algorithm

The greedy algorithm selects the highest-probability cell whenever a new waypoint is required. To prevent excessively long paths, a horizon parameter limits the distance the UAV considers when evaluating cells.

Even with the support of the PGM, the greedy algorithm could lack efficiency, since there's no mechanism penalizing the UAV when it crosses repeated cells multiple times. While this may address the false detection problem, it can significantly hinder the searching process, which led to the implementation of an Artificial Potential Field (APF) based algorithm.

D. Artificial Potential Field algorithm

This approach heavily relies on the original APF [13] but leverages the PGM to improve efficiency. The main idea is to take cells as force sources, creating a force field. The cell with the highest probability will exert an attractive force, whereas the remaining cells will exert a repulsive force. Once again, a horizon parameter is introduced, now to avoid unnecessary interactions and ensure the attractive force remains dominant. The resultant force, combined with a stiffness coefficient k, will determine the displacement vector, as described in (8).

$$\Delta \mathbf{x} = k \sum (\mathbf{F_{rep}} + \mathbf{F_{att}}) \tag{8}$$

The attractive and repulsive forces are computed as follows:

$$\vec{F}_{i,att} = \nabla U_{att} = \begin{cases} \zeta(\vec{x} - \vec{x}_i), & d_i < d_{\min} \\ \zeta \dfrac{\vec{x} - \vec{x}_i}{d(\vec{x}, \vec{x}_i)}, & \text{otherwise} \end{cases} \tag{9}$$

$$\begin{aligned} \vec{F}_{i,rep} &= \nabla U_{rep} \\ &= \begin{cases} 0, & d > d_{\min} \\ \sigma_i \, \eta \Big(\frac{1}{d_{\min}} - \frac{1}{d(\vec{x},\vec{x}_i)} \Big) \frac{(\vec{x}-\vec{x}_i)}{d(\vec{x},\vec{x}_i)^2}, & \text{otherwise} \end{cases} \end{aligned} \tag{10}$$

Note that the attractive potential field is hybrid: quadratic near the goal and conical beyond a threshold distance to enforce force limits. The variables η and ζ represent the potential constants, d_i denotes the distance between the UAV position and cell i, and σ_i corresponds to the prior, mapped logarithmically to the interval $[0, 1]$ as defined in (11):

$$\sigma_i = \frac{-\log p(x_i \mid z_i) - \min_j \big(-\log p(x_j \mid z_j) \big)}{\max_j \big(-\log p(x_j \mid z_j) \big) - \min_j \big(-\log p(x_j \mid z_j) \big)} \tag{11}$$

This formulation successfully assigns lower weights to more likely cells and higher weights to less likely ones. To clarify the dependence of the force on the posterior, when a cell is visited, the update step decreases its likelihood, which in turn increases the corresponding repulsive force. Over time, the prediction step propagates uncertainty along the movement direction. If a visited cell is close to the line extending from the most likely cell in the movement direction, its probability gradually increases, further augmenting the repulsive force.

Two main analyses, from this point onward designated A and B, were performed to evaluate the planners. Unless otherwise noted, the probabilistic grid map (PGM) parameters were defined as follows in Table I.

TABLE I: Simulation parameters for the Probabilistic Grid Map (PGM) and the Artificial Potential Field (APF) path planner

PGM parameters		**APF Parameters**	
UAV altitude	7 m	ζ	100
PGM size	120×120 m	η	80
Cell resolution	2 m	k	$\frac{1}{70}$
Update rate	0.5 s	d_{min}	2 m
β	0.001	d_{max}	30 m
Initial UAV–AUV offset	56 m (A)	Horizon radius	20 m

TABLE II: Mean ToF and standard deviation with fixed UAV start position and sampled GPS pings across 66 runs.

Planner	$\sigma = 10$		$\sigma = 100$	
	Mean [s]	**Std [s]**	**Mean [s]**	**Std [s]**
Spiral	24.71	3.11	62.04	46.04
Greedy	24.01	1.18	94.05	123.56
APF	31.28	5.04	62.95	50.58

IV. Results

A. ToF analysis

To evaluate the planners under different uncertainty levels, the initial AUV position was kept fixed, and the GPS position was sampled from a Gaussian distribution centered on it, with variances of 10 and 100 representing low and high uncertainty, respectively. Note that this variance does not need to correspond to the prior variance σ_P used in the PGM, as it is only introduced to randomize the testing samples; however, in this study, we set them equal. The procedure was structured as follows:

1) Relocate the UAV to a predefined initial position;
2) Generate a simulated GPS ping (variance = σ^2);
3) Initialize the PGM with a Gaussian prior of variance σ_P^2 (we set $\sigma = \sigma_P$) and centered at the GPS ping;
4) Run the simulation and record the time of flight (ToF);
5) Repeat steps 1–4 for statistical validity.

The procedure was repeated 66 times for each combination of variance (σ^2) and planner type, with the GPS ping location held constant for each run and variance level.

Fig. 2 presents the time of flight (ToF, in seconds) as boxplots for the spiral, greedy, and APF planners. Under low-variance (more certain) conditions, the greedy and spiral planners achieve comparable performance, both outperforming the APF planner, whose mean ToF is approximately 30% higher. In contrast, under high-variance conditions, the APF planner performs similarly to the spiral, while the greedy planner shows a clear degradation, with a mean ToF about 50% higher than the others. Table II reports the mean ToF

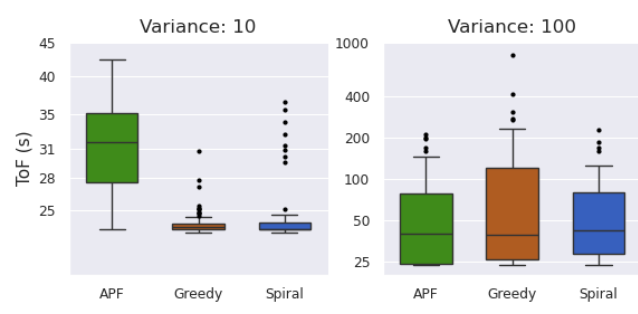

Fig. 2: Boxplots of ToF for each planner and variance value. Each combination of planner type and variance was evaluated over 66 runs.

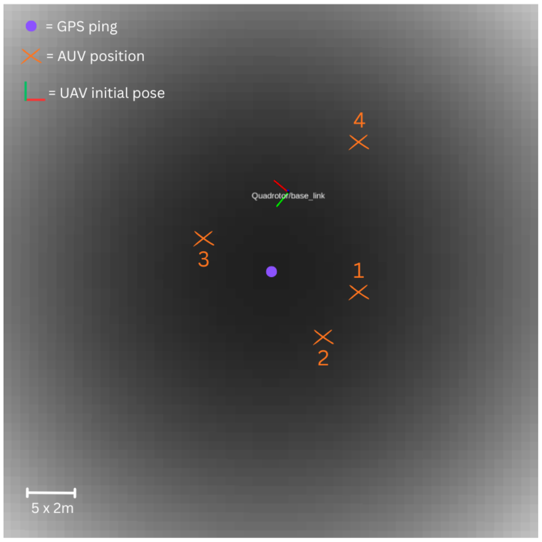

Fig. 3: PGM representation from RVIZ with all AUV initial positions considered in the coverage analysis and the initial UAV pose.

and corresponding standard deviation, which further support these observations.

B. Coverage analysis

In the second experiment, four distinct initial positions of the AUV were tested while keeping the GPS ping fixed. They were not randomly sampled from a distribution because we wanted to evaluate coverage performance for different initial distances between the AUV position and the GPS ping. Under this restriction, they were as random as possible.

Since the algorithms are essentially deterministic, the raw results from each run (*i.e.*, for each combination of AUV position and planner mode) were used directly. The grid map parameters were kept constant, with the prior variance σ_P set to 10. Although the specific value is not critical, maintaining a fixed prior variance ensures that the PGM exhibits a consistent level of uncertainty across all runs. The selected initial positions are shown in Fig. 3, where the GPS measurement is centered in the map and the color intensity

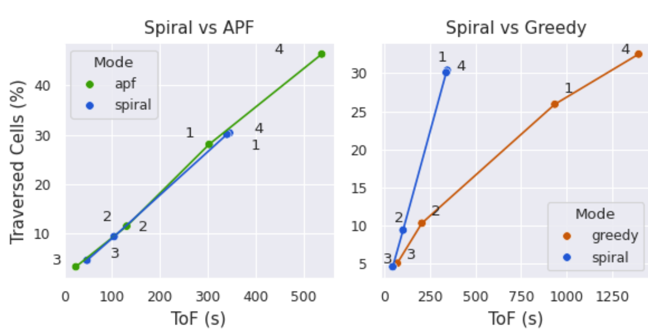

Fig. 4: Percentage of cells traversed (PoCT) versus time of flight (ToF) for each AUV initial position, comparing APF and greedy planners to the spiral baseline.

TABLE III: TpC values for each planner and point.

Planner	Point 1	Point 2	Point 3	Point 4
Spiral	1.83	1.83	1.73	1.84
Greedy	5.44	2.97	2.16	6.48
APF	1.66	1.74	1.23	1.78

indicates the probability of each cell (darker cells correspond to higher probability).

The ToF, and more importantly, both the percentage of traversed cells of the map (PoCT) and the average (mean) number of traversals per seen cell (TpC) were recorded. These metrics were used to assess whether the informative planners effectively mitigate false detections by revisiting high-probability cells while maintaining time efficiency and are presented in Fig. 4 and Table III.

To provide context for interpreting the results, we can qualitatively categorize them into three cases:

1) High ToF: indicative of an inefficient planner (e.g., greedy).
2) Low TpC and low ToF: indicative of an efficient planner (e.g., spiral and APF with our selected parameters).
3) High TpC and low ToF: indicative of a robust planner that recognizes the AUV's likely location; occasional false detections are treated as spurious, preventing extensive deviation from the region of interest (e.g., APF with alternative parameters).

The results indicate that the greedy planner tends to revisit individual cells more frequently, as reflected by the higher TpC values in Table III, without substantially increasing the overall area covered (Fig. 4). Consequently, this results in a longer total ToF compared to the spiral and APF planners. The APF planner, on the other hand, exhibits slightly lower TpC values than the spiral planner (on average, 0.2 fewer traversals per visited cell) while achieving comparable PoCT, except at point 4, where it shows a higher ToF. These observations suggest that, for the parameters that were used, the APF performance is similar to the spiral planner, whereas the greedy planner is clearly inefficient.

V. Conclusions and Future Work

This work introduced a probabilistic framework for a search/coverage problem that integrates a Bayes filter with classical planners (greedy and APF-based). The approach demonstrated that the performance of a deterministic spiral planner can be matched or exceeded under uncertain localization. The findings indicate that the greedy method performs well when GPS measurements are accurate and robustness against false detections is prioritized, though it does so at the cost of efficiency. The APF-based planner, in contrast, achieves a balance between exploration efficiency and adaptability, rivalling the spiral in coverage while leveraging probabilistic reasoning to revisit uncertain regions as needed.

At first glance, the APF results may suggest a deviation from the intended design, which focuses on increasing traversals in high-probability regions. However, balancing exploration efficiency and robustness to false detections depends on careful parameter tuning. In fact, simulation results suggest that the ratio $\frac{\zeta}{\eta}$ strongly influences APF behavior. For higher values, the attractive force dominates, and the APF path closely resembles that of the greedy planner. For lower values, repulsive forces become more significant, and the APF trajectory approximates the spiral planner's behavior. This demonstrates that tuning $\frac{\zeta}{\eta}$ allows controlling the planner's balance between exploration and exploitation.

Furthermore, the planner is also sensitive, for instance, to prior variance, cell resolution, and small changes in the AUV's position, so these results reflect its behavior under the specific experimental conditions rather than its full capabilities.

This study demonstrates the potential of probabilistic CPP for adaptive search tasks, but future work should incorporate more realistic motion and perception models. In particular, a comparison between our convolution-based motion model and a particle filter implementation would be informative. Integrating a perception model is especially important, as it would enable optimization of other parameters (e.g., UAV altitude) and improve the accuracy of the probabilistic grid map. Further investigations should assess whether a binary detection model suffices or if including both $p(\tilde{x}|z)$ and $p(x|z)$ in the probabilistic framework offers advantages, consider more complex priors, and conduct real-world experiments to evaluate robustness beyond simulation. Finally, although energy consumption was not considered here (since mid-mission recharging is possible, though suboptimal), incorporating it into the motion planner via a cost function balancing probabilities and energy could provide valuable insights compared to the current approach.

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